CAN VISION-LANGUAGE MODELS LEARN TO EXECUTE CLASSICAL SEARCH ALGORITHMS?

Anonymous authors
Paper under double-blind review

ABSTRACT

We study whether vision-language models can execute declared classical search algorithms rather than directly generate complete plans. A Search Process Policy emits a Typed Search Operation whose explicit operands determine the next exploration decision. A Trusted Search Runtime validates and applies the emission, maintains bounded external Search Memory, checks Algorithm Invariants, and charges invalid operations without choosing omitted operands or silently repairing outputs. This contract separates stepwise validity from full-episode structural/process competence. In a frozen, outcome-blind 15-task BFS development panel, final process-SFT checkpoints from five training seeds achieved 1.0 invariant-valid success with zero invalid operations; the pretrained base achieved 0.0, while a random-valid reference also achieved 1.0. The learned policy therefore showed executable BFS behavior on the selected panel but no gain over the strongest control, producing a governed `VALID_STOP` with `scientific_completion=false`. The next structural stage uses a complete unpruned BFWS variant: exact traces and a released text-state corpus now exist, but training and adjudication remain incomplete. A*, visual-state, multimodal-state, DAgger, and final comparative evaluation remain prospective.

1 INTRODUCTION

Symbolic planning provides a controlled probe of the reasoning abilities of large models. Tasks have formal specifications and mechanically verifiable candidate answers, and the difficulty of the underlying state space can be varied systematically through domain structure and problem size (Valmeekam et al., 2023; Huang et al., 2026). These properties allow an evaluator to separate what a model can do from what fluent output merely suggests.

This paper extends a prior observational study of latent planning competence in off-the-shelf models (Huang et al., 2026). That study distinguished two competencies. Operational reasoning covers local applicability, progression, and validation. Structural enumeration covers reachability, landmarks, and other global state-space structure. Gains in one did not produce gains in the other. Scaling, chain-of-thought prompting, targeted prompting, and scratchpad or agent harnesses improved operational reasoning, while structural enumeration remained comparatively flat. These findings were item-level and text-only, and they were obtained from models used as released, without task-specific training.

Structural enumeration is closely tied to global search-space reasoning, but it does not amount to complete search capability (Huang et al., 2026). Identifying which states are reachable, or which states serve as landmarks, does not require controlling a frontier across a budgeted episode. Its flat profile under scaling and prompting nonetheless motivates a training-based intervention, because the prior work tested neither acquisition through training nor performance over full budgeted search episodes (Huang et al., 2026). This paper tests whether directly learning to execute a declared classical search algorithm can produce model-attributable structural/process competence beyond valid but uninformed exploration.

A valid local transition does not establish algorithm execution. Classical search algorithms expose decisions that can be checked mechanically: which state is selected, which successor is generated, how frontier and duplicate records change, and whether the declared ordering rule is respected.

Stepwise validity is narrower than algorithmic correctness, and it does not by itself establish termination, completeness, optimality, or goal achievement under a budget. A system that proposes a complete plan directly, or predicts one locally applicable transition, is therefore not necessarily executing breadth-first search (BFS), best-first width search (BFWS), A*, or any other declared procedure. A model can be accurate on applicability, progression, validation, and transition/action tasks while failing to control the exploration decisions that determine a full search episode. We call the former operational competence and the latter full-episode structural/process competence.

Several research threads approach parts of this problem. Classical planning systems place search control in a deterministic algorithm *[TODO: cite]*. Neural algorithm execution studies whether learned models can reproduce intermediate computation and generalize beyond training sizes *[TODO: cite]*. LLM and VLM planning work evaluates plans or actions proposed from text and images *[TODO: cite]*, while tool- and verifier-mediated systems execute model outputs under an external checker *[TODO: cite]*. These lines provide relevant mechanisms and counterexamples. They do not justify a universal novelty claim, and they leave the training-based acquisition question open.

We define the division between model-owned decisions and runtime-owned bookkeeping through an ownership contract. The Search Process Policy must emit the operation type and every operand that changes exploration, including the selected source or frontier state and any successor, action, insertion, retirement, or goal-test decision required by that operation. The Trusted Search Runtime may parse, validate, apply, persist, and reject the emission. It may not select a missing operand, reorder candidates for the policy, replace an invalid choice, or inject a default operation. Replaying the raw policy emissions against the same task and prior Search Memory must reproduce the explored trace. Search Memory remains an external boundary holding frontier, visited or best-cost records, novelty, priorities, and related algorithm state, so the model is not required to store unbounded bookkeeping internally. An invalid emission is charged to the episode budget without repair. This contract makes the attribution test explicit: the runtime guarantees mechanical validity, while the policy's operands determine where search goes next.

The first governed model result now comes from BFS. Issue #54 trained five process-SFT seeds on the observable v6 corpus and evaluated their final checkpoints under a resource-bounded successor phase. The panel was frozen without model outcomes and contained 15 development tasks, one per domain. Process SFT reached every selected goal with invariant-valid traces and no invalid operations. The base model reached none. This is evidence that the trained policy can emit executable BFS operations under the declared interface on that panel. It is not evidence of a learned search advantage: random-valid also solved every selected task, so the absolute gain and paired-bootstrap lower bound over the best control were both zero. The governed outcome is therefore `VALID_STOP`, not scientific completion. The result must remain bounded to the selected panel, which contains 11 easy, three medium, and one hard task.

The structural successor stage has also changed. Qualification found that independent iterated-width attempts up to width three were not corpus-complete, so the active gate migrated to a complete unpruned `full_bfws_goal_count` variant. The BFWS runtime exposes exact duplicate, novelty, goal-count priority, insertion, and enqueue facts through one Markov-sufficient builder shared by trace materialization and live evaluation. Exact evidence now covers 105 development tasks and 69,019 decisions. The released text corpus contains 69,019 process records and 67,215 operational records, with zero recorded split-overlap, conflicting-target, live/corpus-mismatch, parse, runtime-application, token-budget, or held-out-access counters. These are trace and corpus findings. Issue #59 remains open because the authorized BFWS training run and structural-gate adjudication are incomplete.

The remaining program is prospective. The planned checkpoint conditions are pretrained base, process SFT, and process SFT plus DAgger, evaluated alongside random-valid and exact-classical references. Operational-only SFT is not an authorized comparator. Under the current supervisor budget, each future model/cell configuration receives one training run at seed 17. Repeated rollout or reference seeds, where retained, do not estimate training-seed variance. A* h_max and landmark-count adapters, visual-state and multimodal-state corpora, DAgger, model-generated successors, replication, and transfer remain gated. The modality conditions are intended to be matched projections of one authoritative task and decision record with equal Search Memory capacity, but no modality effect can be claimed before those adapters, corpora, estimands, and evaluations are frozen and executed. Whether learning to execute declared search algorithms improves standard planningbenchmark performance, or transfers to broader reasoning, remains an open question that the current evidence does not address.

This paper currently contributes three bounded items.

- **A replay-determinative policy/runtime interface.** Typed Search Operations include the operands that determine exploration, while the no-repair Trusted Search Runtime validates and applies those choices without supplying missing decisions. Operation validity, invariant compliance, episode success, and learned-to-control gain remain separate measurements.
- **A governed bounded BFS result.** On the frozen 15-task development panel, process SFT achieved 1.0 invariant-valid success and zero invalid operations versus 0.0 for the base, but random-valid also achieved 1.0. The retained `VALID_STOP` demonstrates executable behavior without claiming structural advantage or final efficacy.
- **A BFWS structural-search substrate.** The program provides replay-verified exact BFWS traces, a Markov-sufficient shared model-input contract, and a released leakage-audited text corpus over 105 tasks and 69,019 decisions. BFWS training and all later algorithm and modality conclusions remain pending.

The evidence boundary is therefore mixed rather than empty. The BFS pilot supplies a trained, governed, negative comparative result on a restricted development panel. The BFWS stage supplies exact traces and corpus evidence but no model result. A*, modality, DAgger, final held-out evaluation, replication, and transfer remain unrun. The remainder of the paper reviews the relevant literature, specifies the operation and ownership contracts, documents the governed BFS and BFWS evidence, and defines the gates required before broader claims can be made.

References

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A Appendix

The Appendix will retain the complete evidence contract, including operation schemas, Algorithm Invariant definitions, runtime pseudocode, data and split procedures, receipt tables, statistics, adapters, and narrowly scoped feasibility details of retained infrastructure.